Ensemble-Based Spatial Snow Water Equivalent Estimates Using the SWE Evaluation Tool for the Rio Grande Basin

April 05, 2026

Project Team:
Max Gersh[1], Logan Stephenson[2], Ross Palomaki[2], and Karl Rittger[2]

Contributor:
Noah Molotch[2]

[1] New Mexico Office of the State Engineer (NM OSE) Hydrology Bureau
[2] Institute of Arctic and Alpine Research (INSTAAR), University of Colorado Boulder
Work funded by: NASA Western Water Actions Office (WWAO)
Contact: Max.Gersh@ose.nm.gov

Introduction

No spaceborne sensor can directly measure snow water equivalent (SWE). Therefore, the spatial distribution of SWE must be estimated from remotely sensed observations combined with modeling approaches. Existing gridded SWE products exhibit documented accuracy limitations across both mountainous and non-mountainous regions, resulting in a wide range of SWE estimates for any given location. To address this uncertainty, the New Mexico Office of the State Engineer (NM OSE), in collaboration with the Institute of Arctic and Alpine Research (INSTAAR), developed the SWE Evaluation Tool (SWEET). SWEET integrates multiple suitable SWE data products into an ensemble framework, with the objective of producing near-real-time spatial SWE estimates for the Rio Grande Basin that are, on average, as accurate as or more accurate than any individual product alone.

Figure 1 shows the near-real-time (April 05, 2026) ensemble SWE estimates for the Rio Grande Basin. Subsequent sections of this report present detailed ensemble-based SWE maps, along with summaries by subbasin and elevation band. In addition to the ensemble results highlighted in this report, figures and tables for each individual SWE product are also provided for download to support transparency and further analysis. These data sets and visualizations are available on the GitHub repository.

For ease of navigation, hyperlinks are provided throughout this report to help readers quickly access basin-specific sections and supporting data tables.

Rio Grande Basin SWE Overview

Rio Grande Headwaters

Upper Rio Grande

Rio Grande-Elephant Butte (Middle Rio Grande)

About SWEET

Rio Grande Basin SWE Overview

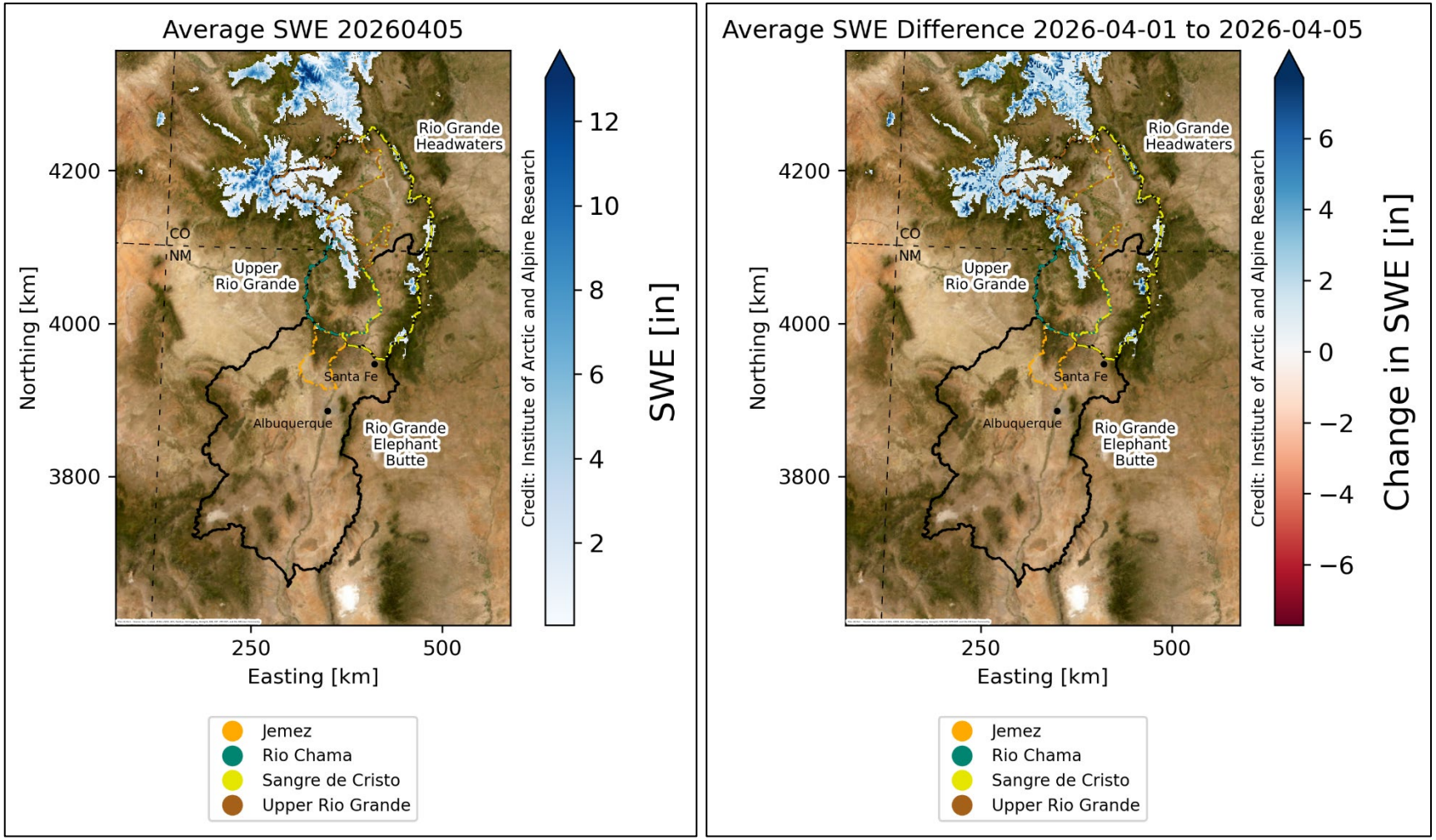

Figure 1. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Basin for April 05, 2026. *SWE amounts (in.) across the Rio Grande Basin (left); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Basin from April 01, 2026, to April 05, 2026 (right); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts.*

Table 1. Estimated ensemble SWE by HUC6 Basin for April 05, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since April 01, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each HUC8 basin (bold).*

HUC6 Basin	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-04-01 (af)	Median Vol. Change since 2026-04-01 (af)
Rio Grande Headwaters	Rio Grande Headwaters	110,997	120,471	74,380 (CU_SWE)	138,140 (UA_SWE)	56,283	67,083
	Alamosa-Trinchera	30,312	31,032	24,880 (CU_SWE)	35,023 (UA_SWE)	20,056	19,582
	San Luis	15,348	14,642	14,475 (CU_SWE)	16,925 (UA_SWE)	4,613	3,621
	Saguache	6,795	6,392	5,101 (SNODAS)	8,890 (UA_SWE)	4,463	4,876
	Conejos	48,426	50,273	41,720 (CU_SWE)	53,285 (UA_SWE)	28,305	34,666
	Total	**211,877**	**221,519**	**161,847 (CU_SWE)**	**252,265 (UA_SWE)**	**113,721**	**115,089**
Upper Rio Grande	Upper Rio Grande	28,022	27,669	7,549 (CU_SWE)	48,848 (SNODAS)	17,709	22,730
	Rio Chama	34,545	39,160	18,353 (CU_SWE)	46,121 (SNODAS)	23,360	28,054
	Total	**62,567**	**66,829**	**25,902 (CU_SWE)**	**94,969 (SNODAS)**	**41,069**	**50,784**
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	233	249	55 (CU_SWE)	396 (SNODAS)	231	249
	Jemez	417	274	36 (CU_SWE)	940 (UA_SWE)	417	274
	Rio Grande-Albuquerque	0	0	0	0	0	0
	Arroyo Chico	0	0	0	0	0	0
	North Plains	0	0	0	0	0	0
	Rio San Jose	0	0	0 (SNODAS, CU_SWE)	1 (UA_SWE)	0	0
	Plains of San Agustin	0	0	0	0	0	0
	Rio Salado	0	0	0	0	0	0
	Jornada Del Muerto	0	0	0	0	0	0
	Elephant Butte Reservoir	0	0	0	0	0	0
	Rio Puerco	207	104	0 (CU_SWE)	516 (UA_SWE)	207	104
	Total	**857**	**773**	**91 (CU_SWE)**	**1,706 (UA_SWE)**	**855**	**773**

Table 2. Estimated ensemble SWE by specified subbasin groupings for April 05, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since April 01, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping (bold).*

OSE Grouping	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-04-01 (af)	Median Vol. Change since 2026-04-01 (af)
Upper Rio Grande	Rio Grande Headwaters	110,997	120,471	74,380 (CU_SWE)	138,140 (UA_SWE)	56,283	67,083
	Saguache	6,795	6,392	5,101 (SNODAS)	8,890 (UA_SWE)	4,463	4,876
	Conejos	48,426	50,273	41,720 (CU_SWE)	53,285 (UA_SWE)	28,305	34,666
	Total	**166,218**	**175,845**	**122,492 (CU_SWE)**	**200,316 (UA_SWE)**	**89,051**	**92,980**
Sangre de Cristo	San Luis	15,348	14,642	14,475 (CU_SWE)	16,925 (UA_SWE)	4,613	3,621
	Alamosa-Trinchera	30,312	31,032	24,880 (CU_SWE)	35,023 (UA_SWE)	20,056	19,582
	Upper Rio Grande	28,022	27,669	7,549 (CU_SWE)	48,848 (SNODAS)	17,709	22,730
	Total	**73,681**	**79,618**	**46,904 (CU_SWE)**	**94,522 (SNODAS)**	**42,379**	**46,811**
Rio Chama	**Total**	**34,545**	**39,160**	**18,353 (CU_SWE)**	**46,121 (SNODAS)**	**23,360**	**28,054**
Jemez	**Total**	**417**	**274**	**36 (CU_SWE)**	**940 (UA_SWE)**	**417**	**274**

Upper Rio Grande Snowpack Update Application

The OSE/ISC Upper Rio Grande Snowpack Update Application (Application) provides water managers and users with detailed SNOTEL-based SWE data organized by the same subbasin groupings shown in Table 2. The Application is updated regularly from December through May and compiles key SWE information from SNOTEL sites within each subbasin. In addition to the Upper Rio Grande groupings (Upper Rio Grande, Sangre de Cristo, Rio Chama, and Jemez), the Application also includes the San Juan River Basin, with data summarized for the Upper San Juan, Animas, and Piedra subbasins.

The Application can be viewed here[1].

[1] Please note that access to the Application may be restricted to employees of the NM OSE/ISC at this time. We are currently working on public access.

Rio Grande Headwaters

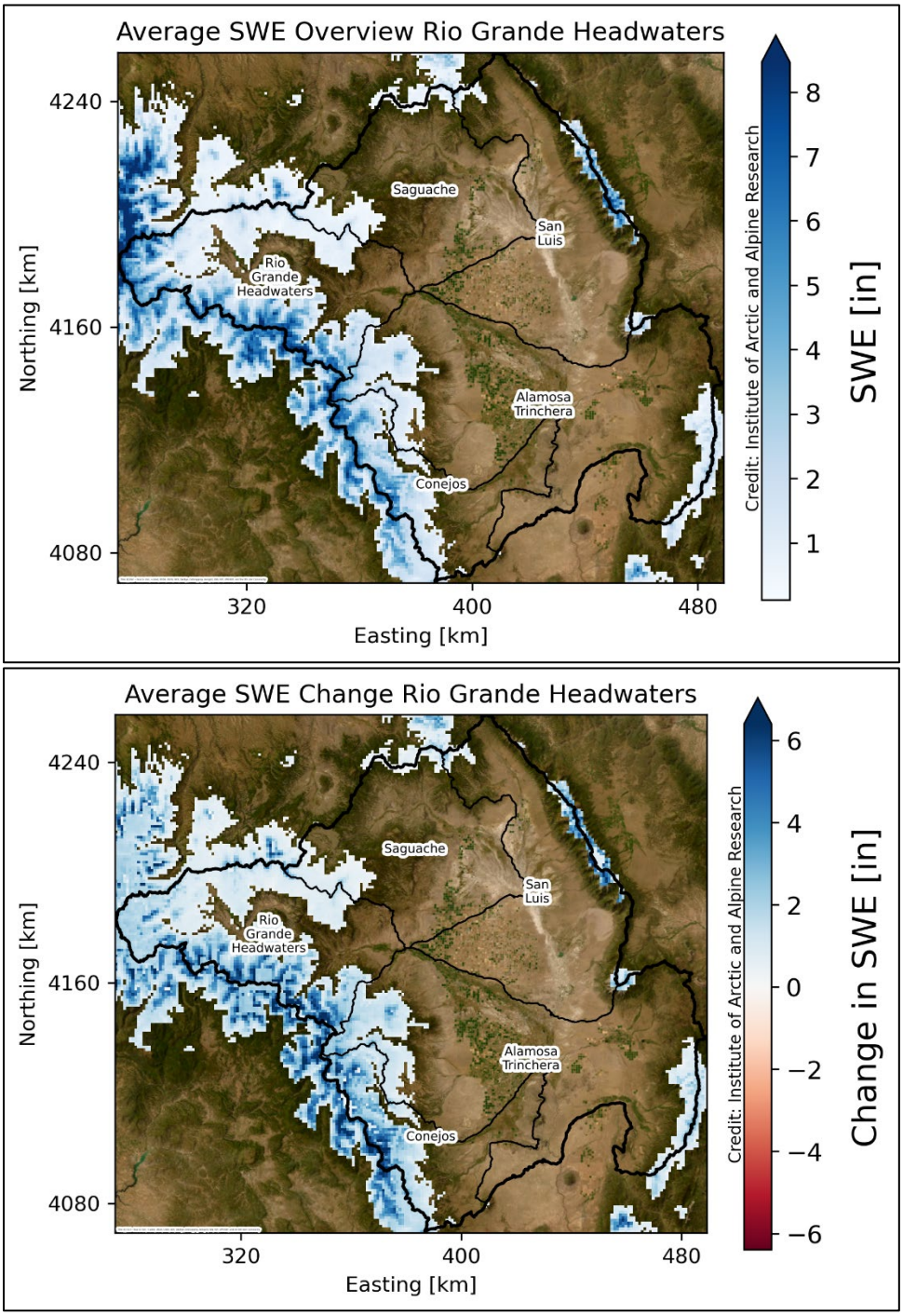

Figure 2. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Headwaters HUC6 Basin for April 05, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande Headwaters Basin, showing spatial patterns of SWE across subbasins including Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Headwaters from April 01, 2026, to April 05, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

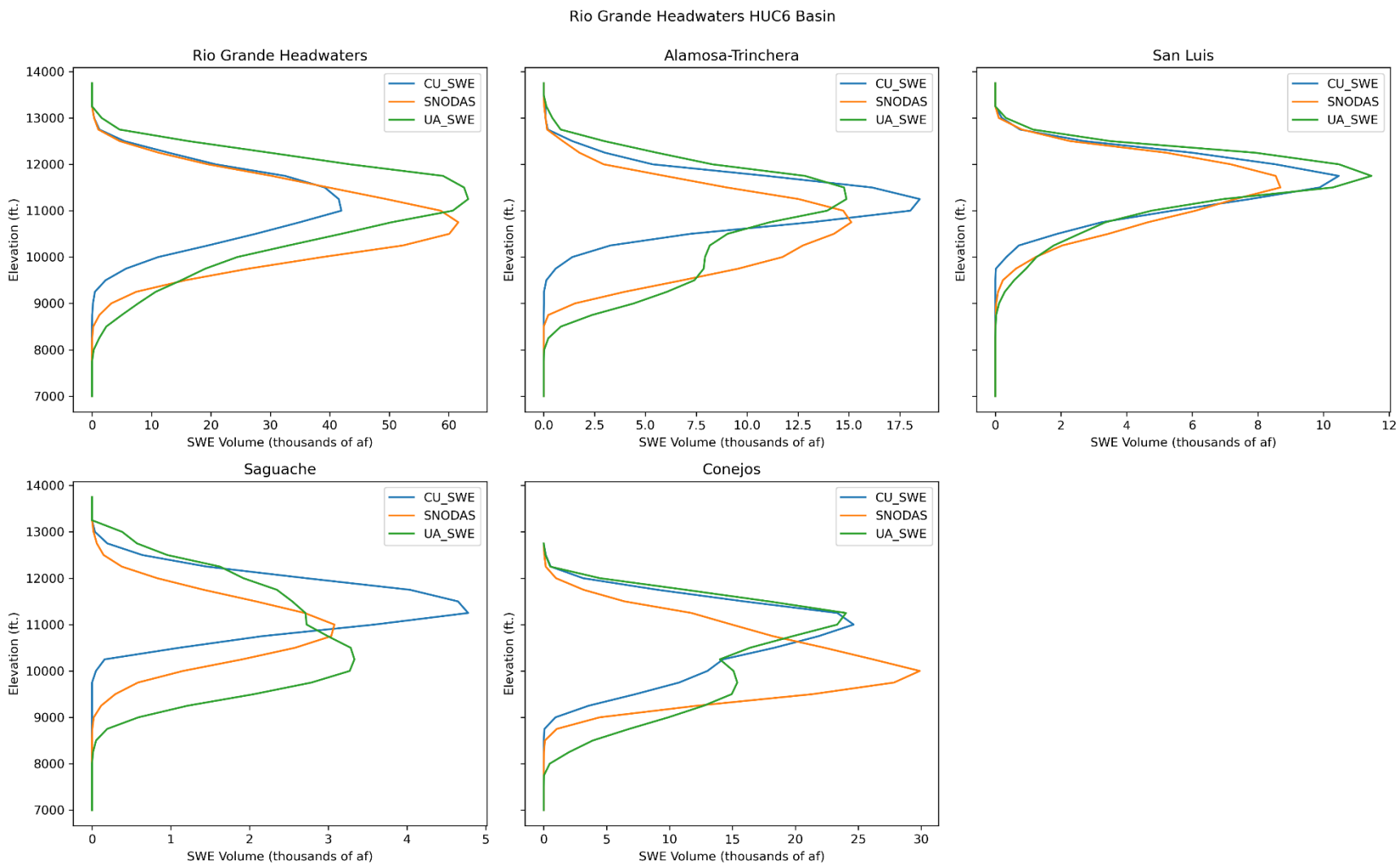

Figure 3. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for April 05, 2026. *Elevation-distributed SWE volume for the Rio Grande Headwaters HUC6 Basin and associated subbasins (Rio Grande Headwaters, Alamosa–Trinchera, San Luis, Saguache, and Conejos), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 3. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for April 05, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since April 01, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-04-01 (af)	Median Vol. Change since 2026-04-01 (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	0	0	0	0	0	0
		8000-9000'	93	0	0 (SNODAS, CU_SWE)	279 (UA_SWE)	93	0
		9000-10000'	3,714	3,227	155 (CU_SWE)	7,758 (UA_SWE)	3,515	3,139
		10000-11000'	24,785	24,437	11,131 (CU_SWE)	38,787 (SNODAS)	17,558	17,736
		11000-12000'	53,756	58,645	41,941 (CU_SWE)	60,683 (UA_SWE)	26,255	28,656
		12000-13000'	27,880	20,798	19,471 (SNODAS)	43,370 (UA_SWE)	8,691	4,644
		13000-14000'	769	354	340 (SNODAS)	1,613 (UA_SWE)	171	63
	Alamosa-Trinchera	7000-8000'	0	0	0	0	0	0
		8000-9000'	8	0	0 (SNODAS, CU_SWE)	24 (UA_SWE)	8	0
		9000-10000'	1,993	1,535	15 (CU_SWE)	4,427 (UA_SWE)	1,988	1,531
		10000-11000'	7,023	7,928	1,406 (CU_SWE)	11,737 (SNODAS)	6,702	7,561
		11000-12000'	15,557	14,720	13,931 (UA_SWE)	18,021 (CU_SWE)	9,168	9,517
		12000-13000'	5,529	5,351	2,948 (SNODAS)	8,287 (UA_SWE)	2,138	1,569
		13000-14000'	202	92	86 (CU_SWE)	426 (UA_SWE)	54	21
	San Luis	7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	47	27	0 (CU_SWE)	114 (UA_SWE)	47	27
		10000-11000'	941	1,231	337 (CU_SWE)	1,255 (UA_SWE)	796	1,073
		11000-12000'	5,422	5,401	4,762 (UA_SWE)	6,104 (SNODAS)	1,916	1,893
		12000-13000'	8,729	8,538	7,176 (SNODAS)	10,472 (UA_SWE)	1,818	2,082
		13000-14000'	209	199	105 (SNODAS)	322 (UA_SWE)	36	39
	Saguache	7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	204	22	0 (CU_SWE)	589 (UA_SWE)	202	22
		10000-11000'	1,492	1,155	49 (CU_SWE)	3,272 (UA_SWE)	1,380	1,079
		11000-12000'	3,129	3,077	2,725 (UA_SWE)	3,586 (CU_SWE)	2,079	2,864
		12000-13000'	1,823	1,922	828 (SNODAS)	2,720 (CU_SWE)	778	921
		13000-14000'	146	37	19 (SNODAS)	382 (UA_SWE)	25	8
	Conejos	7000-8000'	0	0	0	0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-04-01 (af)	Median Vol. Change since 2026-04-01 (af)
Rio Grande Headwaters	Conejos	8000-9000'	160	2	0 (CU_SWE)	479 (UA_SWE)	160	2
		9000-10000'	5,103	4,454	939 (CU_SWE)	9,915 (UA_SWE)	4,770	4,226
		10000-11000'	19,330	15,087	13,015 (CU_SWE)	29,888 (SNODAS)	13,013	7,768
		11000-12000'	20,969	23,326	14,951 (SNODAS)	24,631 (CU_SWE)	9,399	11,573
		12000-13000'	2,864	3,135	979 (SNODAS)	4,479 (UA_SWE)	964	592

Upper Rio Grande

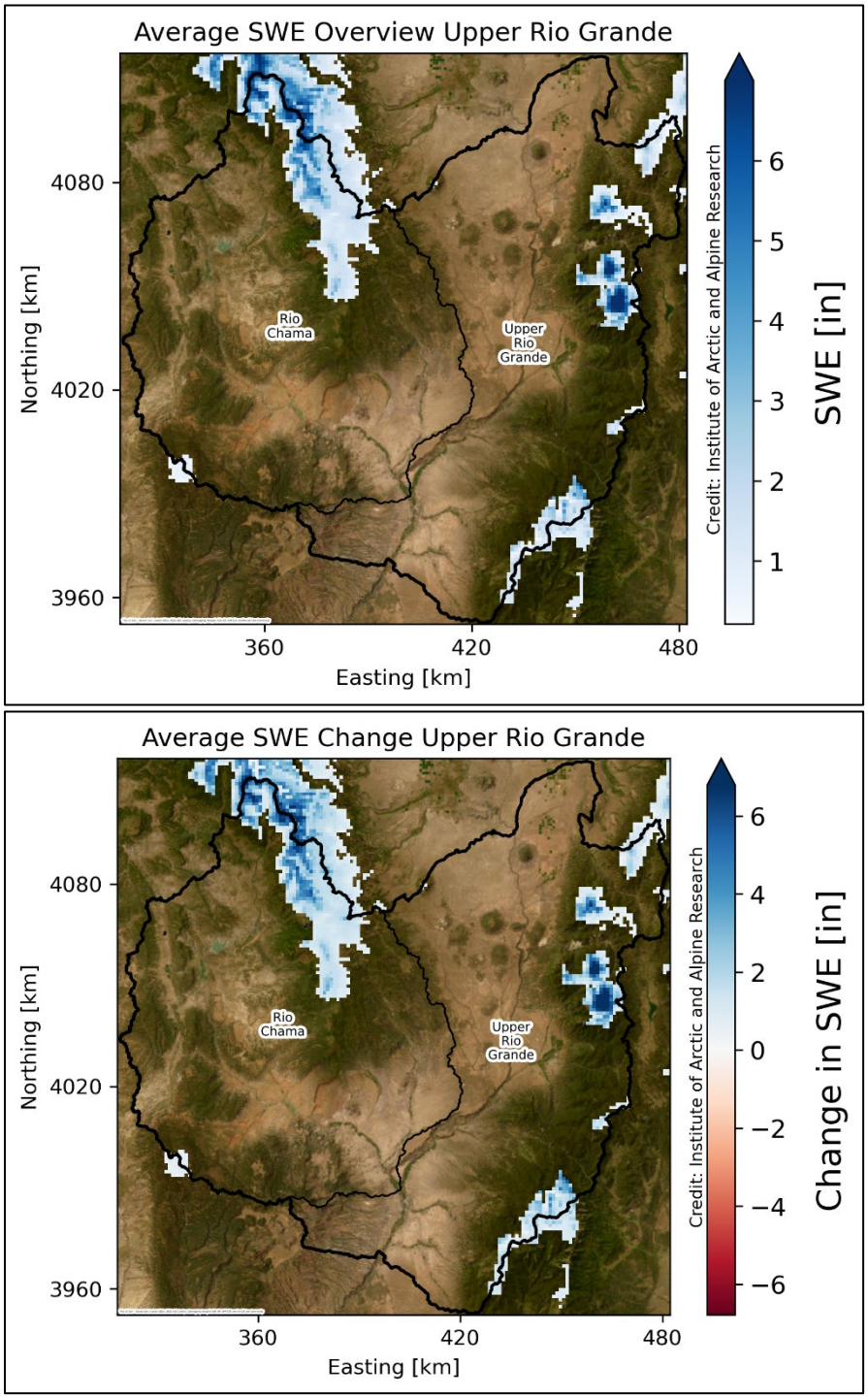

Figure 4. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Upper Rio Grande HUC6 Basin for April 05, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Upper Rio Grande Basin, showing spatial patterns of SWE across subbasins including Rio Chama and Upper Rio Grande (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Upper Rio Grande Basin from April 01, 2026, to April 05, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

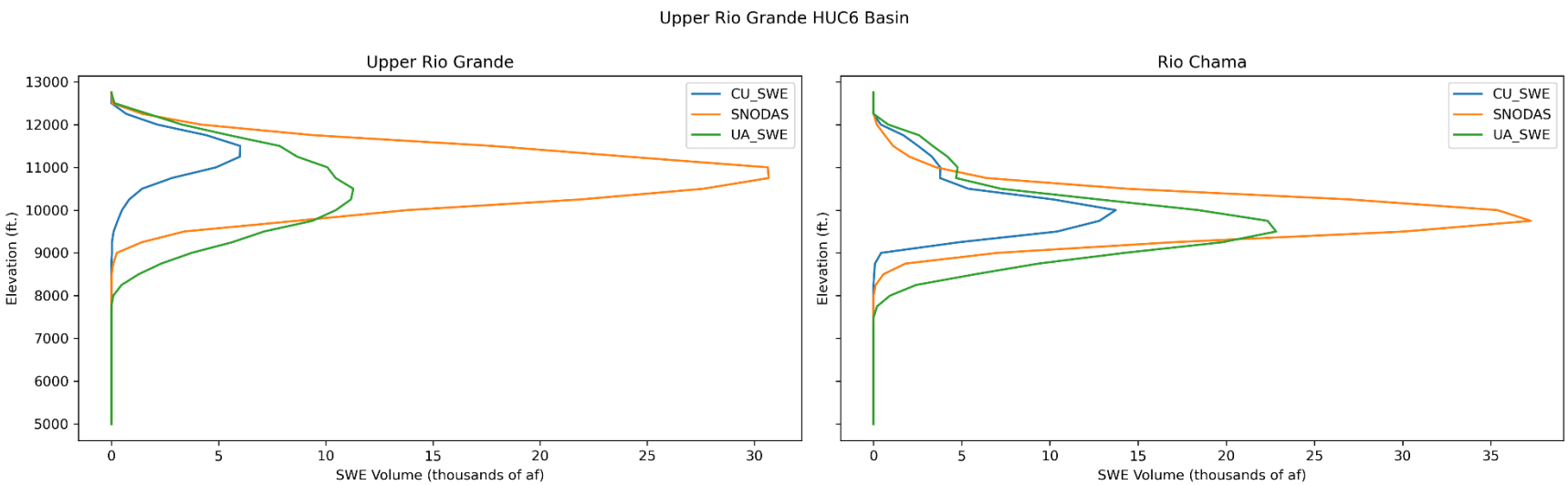

Figure 5. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Upper Rio Grande HUC6 Basin for April 05, 2026. *Elevation-distributed SWE volume for the Upper Rio Grande HUC6 Basin and associated subbasins (Rio Chama and Upper Rio Grande), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 4. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Upper Rio Grande HUC6 Basin for April 05, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since April 01, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-04-01 (af)	Median Vol. Change since 2026-04-01 (af)
Upper Rio Grande	Upper Rio Grande	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	29	0	0 (SNODAS, CU_SWE)	88 (UA_SWE)	29	0
		9000-10000'	1,340	252	28 (CU_SWE)	3,740 (UA_SWE)	1,331	244
		10000-11000'	8,253	10,471	495 (CU_SWE)	13,792 (SNODAS)	6,921	10,318
		11000-12000'	15,198	10,080	4,886 (CU_SWE)	30,628 (SNODAS)	7,973	6,542
		12000-13000'	3,202	3,290	2,140 (CU_SWE)	4,177 (SNODAS)	1,455	1,389
	Rio Chama	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	309	1	0 (CU_SWE)	927 (UA_SWE)	309	1
		9000-10000'	7,215	7,005	430 (CU_SWE)	14,210 (UA_SWE)	6,618	6,860
		10000-11000'	22,509	18,422	13,728 (CU_SWE)	35,377 (SNODAS)	14,602	10,830
		11000-12000'	4,034	3,784	3,549 (SNODAS)	4,768 (UA_SWE)	1,669	1,435
		12000-13000'	478	411	189 (SNODAS)	833 (UA_SWE)	162	77

Rio Grande-Elephant Butte (Middle Rio Grande)

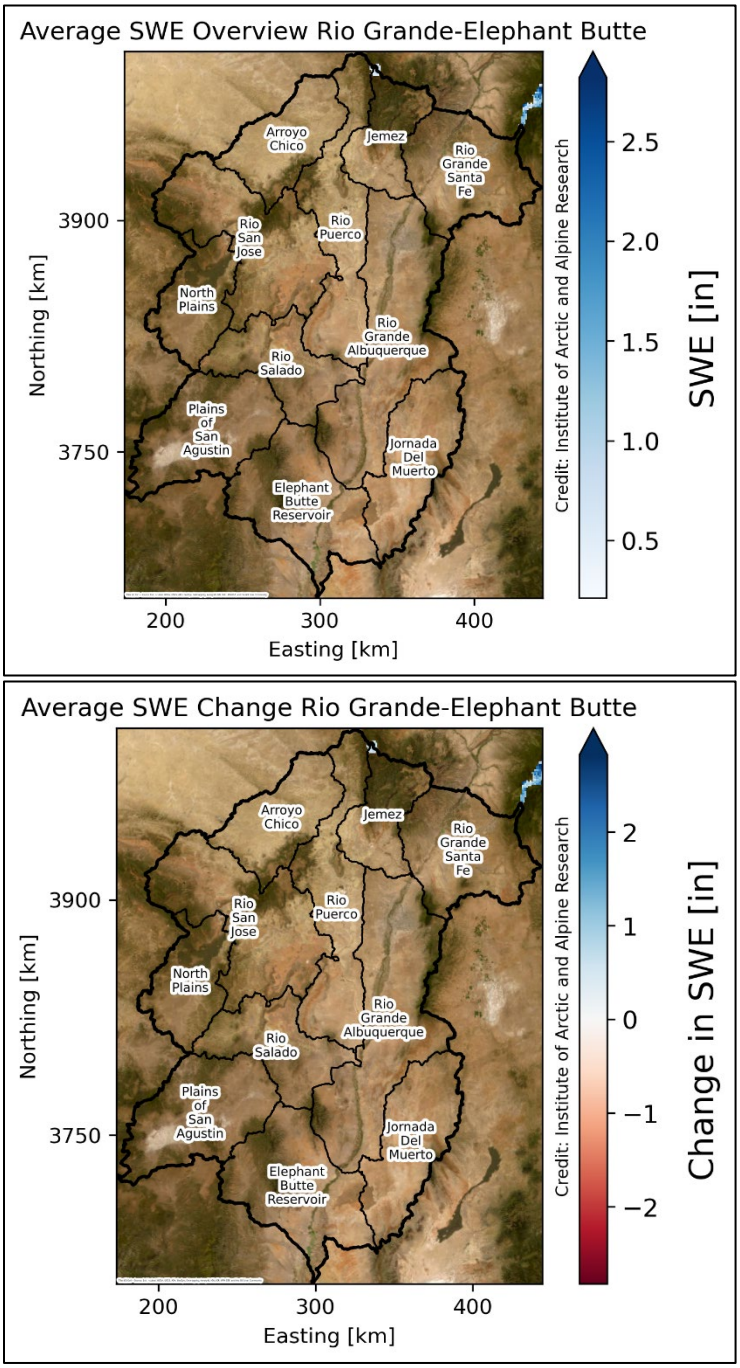

Figure 6. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande-Elephant Butte HUC6 Basin for April 05, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande-Elephant Butte Basin, showing spatial patterns of SWE across subbasins including Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande-Elephant Butte Basin from April 01, 2026, to April 05, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

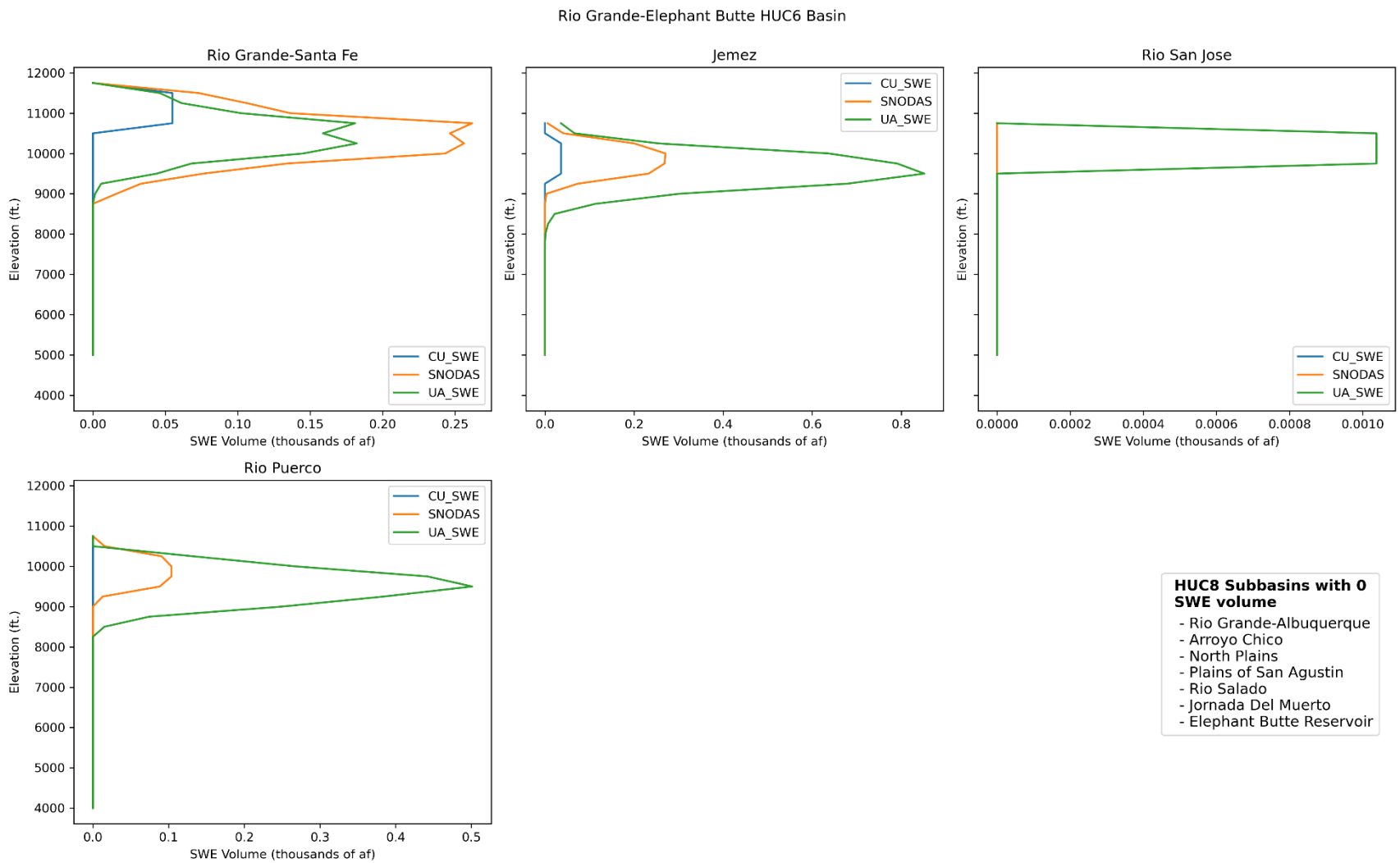

Figure 7. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for April 05, 2026. *Elevation-distributed SWE volume for the Rio Grande-Elephant Butte HUC6 Basin and associated subbasins (Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 5. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for April 05, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since April 01, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-04-01 (af)	Median Vol. Change since 2026-04-01 (af)
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	6	2	0 (CU_SWE)	16 (SNODAS)	6	2
		10000-11000'	129	145	0 (CU_SWE)	243 (SNODAS)	129	145
		11000-12000'	98	102	55 (CU_SWE)	136 (SNODAS)	96	102
	Jemez	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0 (SNODAS, CU_SWE)	1 (UA_SWE)	0	0
		9000-10000'	102	3	0 (CU_SWE)	302 (UA_SWE)	102	3
		10000-11000'	315	271	36 (CU_SWE)	637 (UA_SWE)	315	271
	Rio Grande-Albuquerque	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
		10000-11000'	0	0	0	0	0	0
	Arroyo Chico	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
		10000-11000'	0	0	0	0	0	0
	North Plains	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
	Rio San Jose	5000-6000'	0	0	0	0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-04-01 (af)	Median Vol. Change since 2026-04-01 (af)
Rio Grande-Elephant Butte	Rio San Jose	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
		10000-11000'	0	0	0 (SNODAS, CU_SWE)	1 (UA_SWE)	0	0
	Plains of San Agustin	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
	Rio Salado	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
	Jornada Del Muerto	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
	Elephant Butte Reservoir	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
		10000-11000'	0	0	0	0	0	0
	Rio Puerco	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	83	0	0 (SNODAS, CU_SWE)	249 (UA_SWE)	83	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-04-01 (af)	Median Vol. Change since 2026-04-01 (af)
Rio Grande-Elephant Butte	Rio Puerco	10000-11000'	124	104	0 (CU_SWE)	267 (UA_SWE)	124	104

About SWEET

The following information provides a brief overview of SWEET, and the SWE products currently included in the ensemble framework. For more detailed descriptions, citations, and additional information, please refer to the GitHub repository.

SWE products currently used include:

1. **SNODAS SWE**
    a. The SNOw Data Assimilation System (SNODAS) is a physically based snow model developed by NOAA's National Operational Hydrologic Remote Sensing Center and operational since 2004. SNODAS produces daily 1-km gridded estimates of SWE and related snow variables by combining modeled weather inputs with satellite, airborne, and in situ snow observations.
2. **UA/SWANN SWE**
    a. The University of Arizona/Snow Water Artificial Neural Network Modeling System (UA/SWANN) is a research SWE product that uses snow models, artificial neural networks, and assimilated in situ measurements to generate daily gridded SWE and snow depth. The dataset provides 4-km and downscaled 800-m SWE across the conterminous United States, incorporating SNOTEL and COOP observations with PRISM precipitation and temperature fields.
3. **CU SWE**
    a. The University of Colorado (CU) SWE-Fusion product is an operational dataset produced by INSTAAR that provides near–real-time SWE estimates at 500-m resolution. The approach fuses SNOTEL observations with historical SWE reanalyses and terrain variables using a generalized linear model, and scales results using satellite-derived fractional snow-covered area from Snow Today.

The following notes describe key processing methods, assumptions, and visualization conventions used in generating SWE Evaluation Tool figures, tables, and elevation plots.

Figures:

- For ensemble figures, CU SWE and UA SWE are reprojected to the SNODAS resolution (1 sq km) and averages are only taken at pixels where at least 2 of the 3 products have SWE values greater than 0 inches.
- All overview and subbasin plots set pixels with values of 0 inches to NA so that the basemap is visible.
- For individual product figures, each product is displayed at its native resolution.
- Elevation bands of 250 feet are used in the elevation figures, and SWE volume is calculated for each product's native resolution in each band.
- Difference plots display pixels where snow was present in either the current or previous report.
- Figures are available in PNG format on the GitHub repository.

Tables:

- For all tables, mean, median, and volume are calculated at each product's native resolution - zeros are not removed from mean and median calculations.
- For ensemble tables, mean and median volume are calculated using the total volumes from the 3 products - not from an average aggregate raster.
- For min. and max., if 2 products have 0 volume, both products are listed in that cell. If all 3 products have 0 volume, no product is listed for min. or max. in that row.
- Change in SWE will only be calculated starting with the second report of the water year.
- Ensemble calculations: For HUC8 subbasins, the ensemble (mean and median) SWE and volume change (mean and median) are calculated individually for each subbasin. For the HUC6 "Total" row (tables 1 and 2), the ensemble is recalculated using the full HUC6 basin extent rather than summing HUC8. While the ensemble mean is additive and therefore consistent between HUC8 sums and the HUC6 total, the median reflects the middle value among the three products for the entire basin. As a result, median values will not sum to the HUC6 total, since they represent the difference between basin-wide medians rather than the sum of subbasin medians.
- Tables are available in CSV and PDF format on the GitHub repository.
- Elevation tables presented in the report (Tables 3, 4, and 5) are combined into one table (Table 3) on the GitHub repository.

Acknowledgments

The work completed and presented in this report is funded by NASA Western Water Actions Office (WWAO), subcontract no. 1712706, as part of the *Rio Grande River Basin Needs Assessment Report*. The SWE Evaluation Tool is a collaborative effort from members at INSTAAR, University of Colorado Boulder, and the New Mexico Office of the State Engineer.

We would also like to extend our appreciation to Noah Molotch and the Mountain Hydrology Group at CU Boulder, INSTAAR, for granting our team access to the CU SWE operational SWE dataset.